

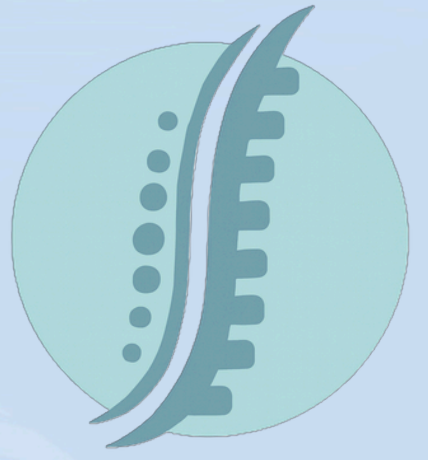
WELLALIGN



UG- Final Year Project

AI | Posture | Computer Vision | Machine Learning | Physical Wellbeing

AI-POWERED POSTURE CORRECTION AND EXERCISE RECOMMENDATION APPLICATION FOR PHYSICAL WELLBEING



Problem Statement :

Postural health is often overlooked in the society. Poor posture in desk-based workers and students causes musculoskeletal strain and reduced wellbeing. Existing tools focus only on classification performance or provide limited real-world guidance. There is a gap between isolated posture predictions and end-to-end support workflows that detect issues, explain findings, recommend actions, and enable longitudinal progress tracking.

Key Features :

- ✓ **Image-based Assessment:** upload front/side posture photos → ML analysis → scoring
- ✓ **Real-time Monitoring:** live webcam tracking → issue detection → session summaries
- ✓ **AI Prescription:** Gemini-powered guidance → exercise recommendations → video links
- ✓ **Progress Tracking:** historical assessments/sessions → trend charts → personalisation
- ✓ **Safety-First:** medical disclaimers, non-diagnostic framing

Evaluation :

Model Performance:

- Validation Accuracy: 0.98 | Macro F1: 0.97 | Weighted F1: 0.98

System Testing:

- End-to-end workflows reliable under normal and failure conditions

Feature Utility:

- AI enrichment improves feedback interpretability and actionability
- Historical persistence enables longitudinal trend analysis and personalisation

Project Author & Supervisor :

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Technology Used:

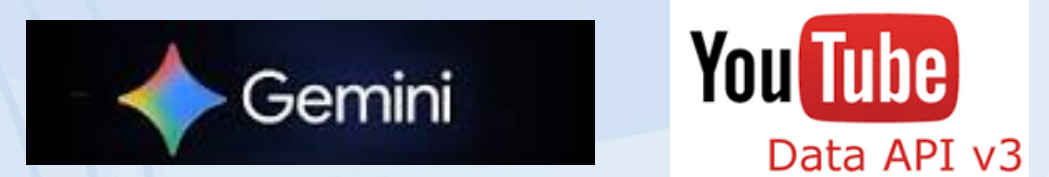
Frontend:



Backend:



External APIs:



ML Service:



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